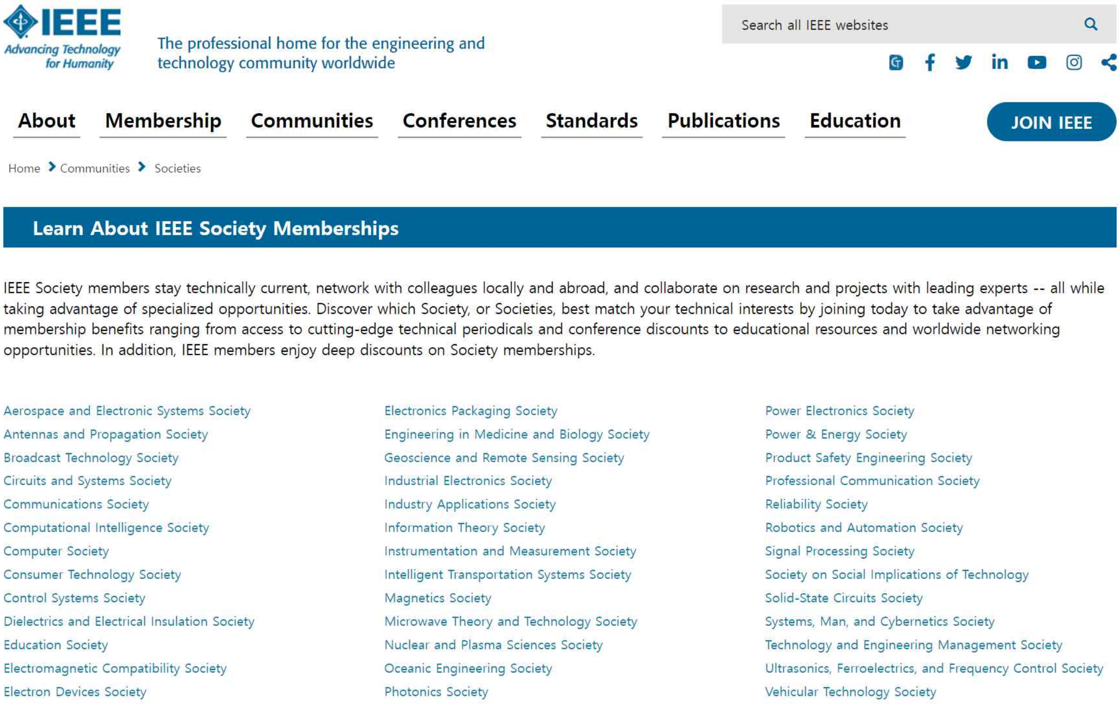
Electrical and Computer Engineering: An Overview

IEEE : Institute of Electrical and Electronics Engineers

https://www.ieee.org/


IEEE Advancing Technology for Humanity

The professional home for the engineering and technology community worldwide

Search all IEEE websites

About | Membership | Communities | Conferences | Standards | Publications | Education | JOIN IEEE

Home > Communities > Societies

Learn About IEEE Society Memberships

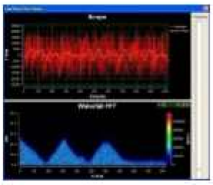
IEEE Society members stay technically current, network with colleagues locally and abroad, and collaborate on research and projects with leading experts -- all while taking advantage of specialized opportunities. Discover which Society, or Societies, best match your technical interests by joining today to take advantage of membership benefits ranging from access to cutting-edge technical periodicals and conference discounts to educational resources and worldwide networking opportunities. In addition, IEEE members enjoy deep discounts on Society memberships.

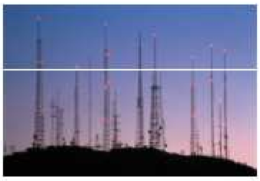
Aerospace and Electronic Systems Society
Antennas and Propagation Society
Broadcast Technology Society
Circuits and Systems Society
Communications Society
Computational Intelligence Society
Computer Society
Consumer Technology Society
Control Systems Society
Dielectrics and Electrical Insulation Society
Education Society
Electromagnetic Compatibility Society
Electron Devices Society
Electronics Packaging Society
Engineering in Medicine and Biology Society
Geoscience and Remote Sensing Society
Industrial Electronics Society
Industry Applications Society
Information Theory Society
Instrumentation and Measurement Society
Intelligent Transportation Systems Society
Magnetics Society
Microwave Theory and Technology Society
Nuclear and Plasma Sciences Society
Oceanic Engineering Society
Photonics Society
Power Electronics Society
Power & Energy Society
Product Safety Engineering Society
Professional Communication Society
Reliability Society
Robotics and Automation Society
Signal Processing Society
Society on Social Implications of Technology
Solid-State Circuits Society
Systems, Man, and Cybernetics Society
Technology and Engineering Management Society
Ultrasonics, Ferroelectrics, and Frequency Control Society
Vehicular Technology Society

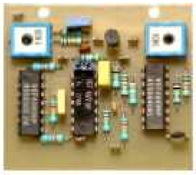
What is Electrical and Computer Engineering?

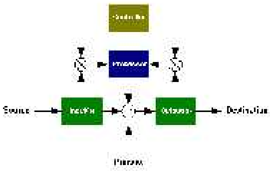
- Branch of engineering on the application of electricity in all its forms.
- It deals with the principles and application of electricity, electronics, and electromagnetism.
- (Usually outside the US) electrical engineering is associated with large-scale electrical systems (power transmission, motor control) whereas electronic engineering is concerned about the small-scale electronic systems (computers and integrated circuits).


Power Electronics


Signal Processing

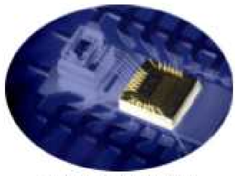

Telecommunication


Electronic Circuit & Microelectronics


Systems & Control

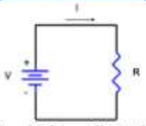

Computer Systems

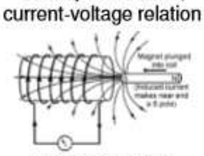

Mechatronics

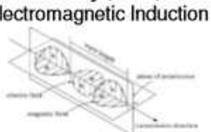

Instrumentation & Sensors

History of Electrical and Computer Engineering

Post-20th Century


Ohm's Law (1827): Concept of resistor, current-voltage relation


Faraday (1831): Electromagnetic Induction


Maxwell Equation (1873): Unified theory of electricity & magnetism

- Coulomb - Charge, electrostatics
- Volta – Voltaic battery, electropile
- Edison – Light bulb, large scale electrical supply network

Early-20th Century


Hertz(1888) & Marconi (1896): Transmission of radio waves


Watson-Watt (1936): Radar


Vacuum tube & ENIAC (1946) & Transistor (1947)

- De Forest – Amplifier tube (triode)
- Armstrong – Electronic TV (1931)
- Randall & Boot – Magnetron (1940)

Late-20th ~ 21st Century


Jack Kilby (1956): First integrated circuit


Altair 8800 (1975) First personal computer


Intel Quad Core (2008)


Samsung LCD TV


Apple iPhone (2007)


Tesla Electric Car

Applications of Electrical, Computer and Mechanical Engineering

[Example 1] Self-driving Car (Autonomous Car)

https://en.wikipedia.org/wiki/Self-driving_car

[Example 2] [Manufacturing systems] Robots in a car maker

[Example 3] Biomedical engineering systems

Fusion of electrical engineering, electronics engineering,
computer engineering, mechanical engineering,
material science, chemistry, medical science, and ......

[Example 4] Airplane

[Example 5] Digital Camera

[Example 6] Smart Phone

※ You must know about Hardwares and Softwares,
if you want to work in those companies.
If you know about only softwares, ...

This course is a basic hardware course, ...

● Contents of Electric Circuit

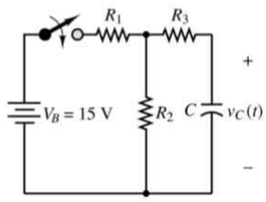
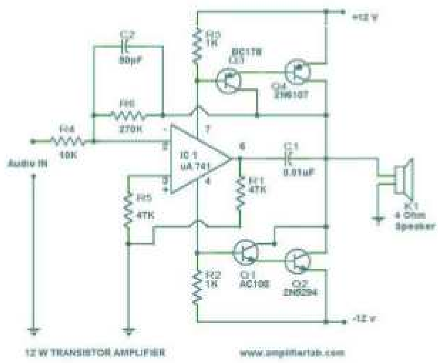
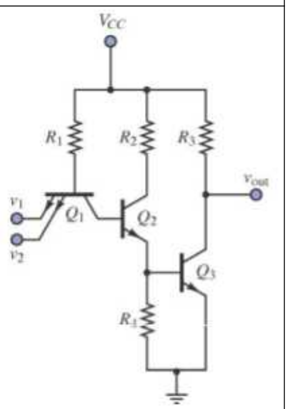
Chap.1 Circuit variables

§1.1 Electric Circuit Theory

- The most basic subjects in Electrical and Computer Engineering
  Electric Circuit Theory [hardware]
  Languages and Data Structure [software]

- An **electric circuit** is a **mathematical model** that approximates the behavior of an actual electrical system.

- We will study how to analyze and design a circuit.

Circuit Examples		
Resistor-Capacitor Transient Circuit	Audio Amplifier	Digital NAND gate
		
differential equation Laplace transform complex number	Laplace transform	Boolean Algebra 2's complement (0/1 : very easy)

※ <u>circuit analysis</u>
Find the voltage and the current of an element in a circuit.

§1.2 The International System of Units (SI)

- The SI units are base on seven defined quantities:

TABLE 1.1 The International System of Units (SI)

Quantity	Basic Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	degree kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

- Derived Units in SI

TABLE 1.2 Derived Units in SI

Quantity	Unit Name (Symbol)	Formula
Frequency	hertz (Hz)	s^{-1}
Force	newton (N)	$kg \cdot m/s^2$
Energy or work	joule (J)	$N \cdot m$
Power	watt (W)	J/s
Electric charge	coulomb (C)	$A \cdot s$
Electric potential	volt (V)	J/C
Electric resistance	ohm (Ω)	V/A
Electric conductance	siemens (S)	A/V
Electric capacitance	farad (F)	C/V
Magnetic flux	weber (Wb)	$V \cdot s$
Inductance	henry (H)	Wb/A

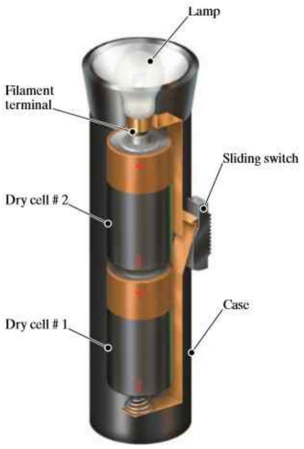
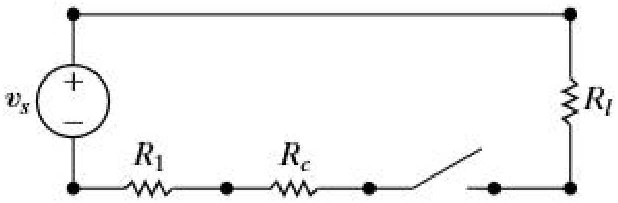
- Standardized Prefixes to Signify Powers of 10

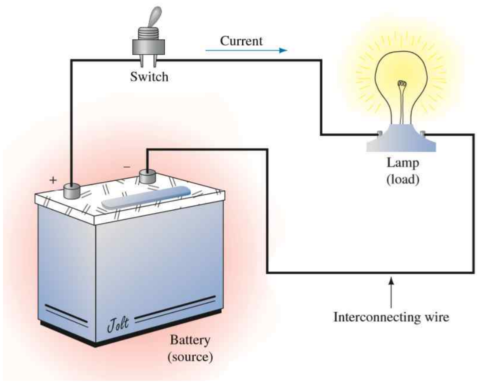
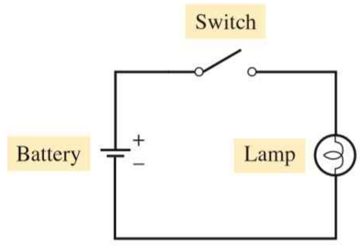
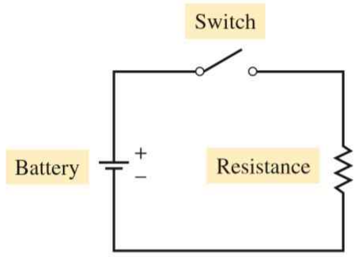
power	symbol	prefix
10^{-1}	d	deci
10^{-2}	c	centi
10^{-3}	m	milli
10^{-6}	μ	micro
10^{-9}	n	nano
10^{-12}	p	pico
10^{-15}	f	femto
10^{-18}	a	atto
10^{-21}	z	zepto
10^{-24}	y	yocto

power	symbol	prefix
10	da	deka
10^{2}	h	hecto
10^{3}	k	kilo
10^{6}	M	mega
10^{9}	G	giga
10^{12}	T	tera
10^{15}	P	peta
10^{18}	E	exa
10^{21}	Z	zetta
10^{24}	Y	yotta

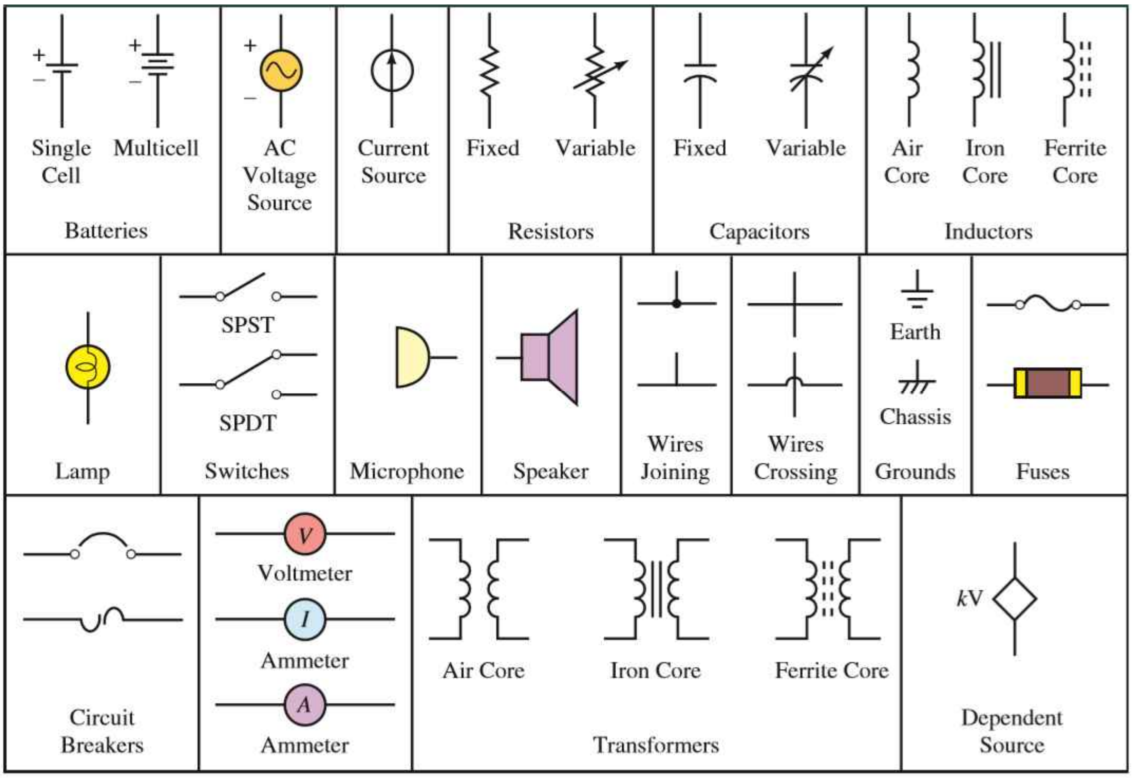
§1.3 Circuit Analysis: An Overview

- circuit models
- ideal circuit components
- circuit analysis

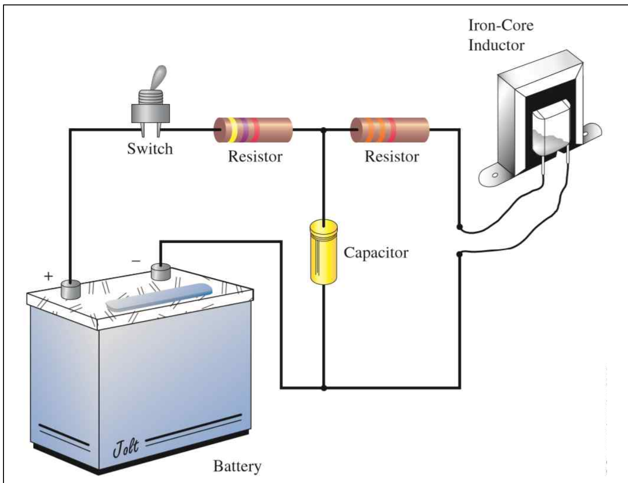
real system		circuit model
	⇒	 ※ We can use mathematics in order to analyze this circuit.

Pictorial diagram (real system)	schematic diagram using lamp symbol
	
	schematic diagram using resistance symbol  (※ We can use mathematics in order to analyze this circuit.)

※ Example of Schematic Circuit Symbols



※ SPST: single pole single throw, SPDT: single pole double throw

	schematic circuit diagram ?

§1.4 Voltage and Current

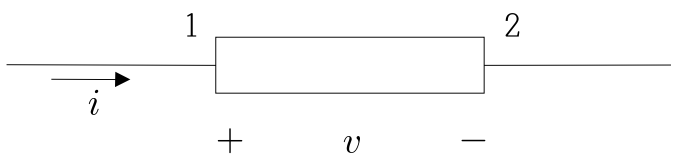
- Voltage and Current defined in Physics

definition of voltage	definition of current
$v = \dfrac{dw}{dq}$	$i = \dfrac{dq}{dt}$
$v =$ the voltage in volts $w =$ the energy in joules $q =$ the charge in coulombs	$i =$ the current in amperes $q =$ the charge in coulombs $t =$ the time in seconds

※ These definitions are not widely used in electrical engineering.

§1.5 The Ideal Basic Circuit Element

- an ideal basic circuit element

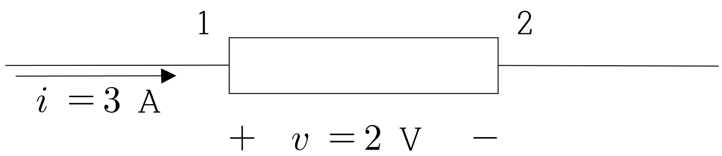


(1) It has only two terminals in circuit theory.

(2) The characteristics of the element is described mathematically in terms of current and/or voltage. (black box model)

※ (v is function of i) or (i is function of v)

(example) Interpretation of reference directions in the above figure



2 volt voltage drop from terminal 1 to terminal 2

or 2 volt voltage rise from terminal 2 to terminal 1

※ Power $p = vi = 2[\mathrm{V}] \times 3[\mathrm{A}] = 6[\mathrm{W}]$ [W: Watt]

6 watts are consumed by the element. (heat)

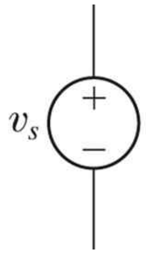
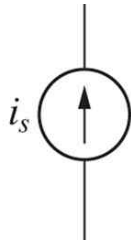
※ $p > 0$: power consumption by the circuit element (example: resistors)

$p < 0$: power generation by the circuit element (example: batteries)

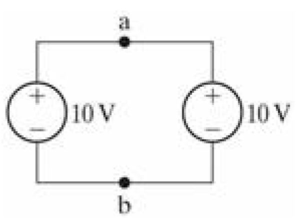
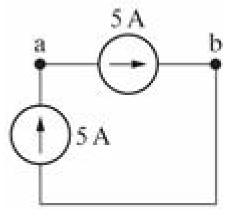
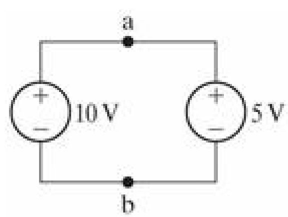
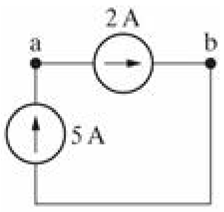
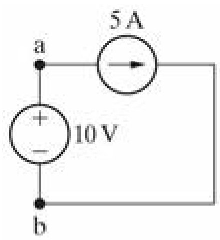
Chap.2 Circuit Elements

§2.1 Voltage Sources and Current Sources

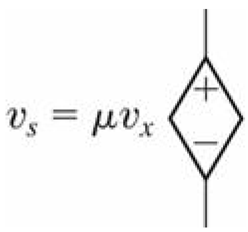
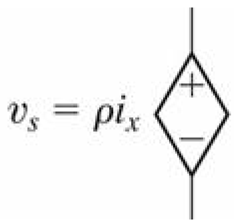
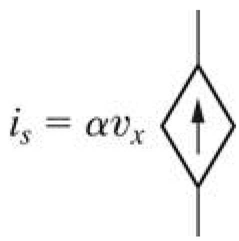
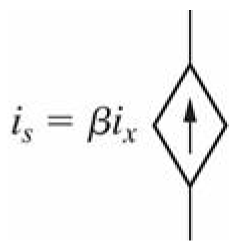
- Ideal Independent Source

ideal independent voltage source	ideal independent current source
	

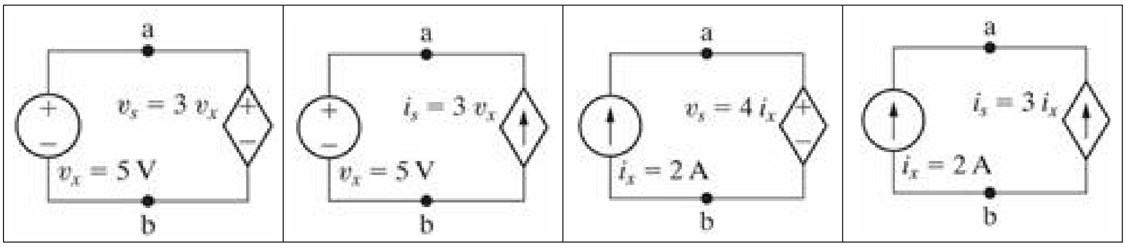
(Example) valid or not?

- Ideal Dependent Source

an ideal dependent voltage-controlled voltage source (VCVS)	an ideal dependent current-controlled voltage source (CCVS)	an ideal dependent voltage-controlled current source (VCCS)	an ideal dependent current-controlled current source (CCCS)
			

(Example) valid or not?



- active elements : can generate electric energy
  passive elements : cannot generate electric energy

(Examples)

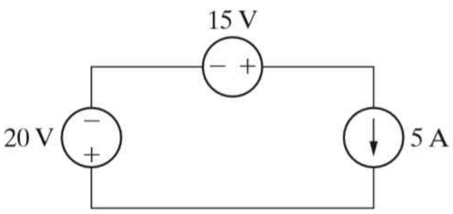
2.1 a) Is the interconnection of ideal sources in the circuit in Fig. P2.1 valid? Explain.

b) Identify which sources are developing power and which sources are absorbing power.

c) Verify that the total power developed in the circuit equals the total power absorbed.

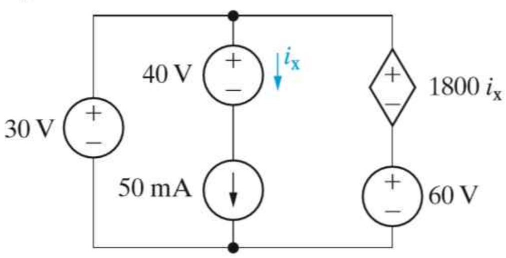
d) Repeat (a)–(c), reversing the polarity of the 20 V source.

Figure P2.1



2.9 If the interconnection in Fig. P2.9 is valid, find the total power developed in the circuit. If the interconnection is not valid, explain why.

Figure P2.9



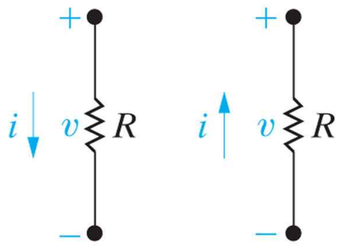
● Batteries (※ Read at home)

Type of battery	Common uses	Hazardous component	Disposal recycling options
Wet cell (i.e. a primary cell that has a liquid electrolyte)			
Lead–acid batteries	Electrical energy supply for vehicles including cars, trucks, boats, tractors and motorcycles. Small sealed lead–acid batteries are used for emergency lighting and uninterruptible power supplies	Sulphuric acid and lead	Recycle – most petrol stations and garages accept old car batteries, and council waste facilities have collection points for lead–acid batteries
Dry cell: Non-chargeable – single use (for example, AA, AAA, C, D, lantern and miniature watch sizes)			
Zinc carbon	Torches, clocks, shavers, radios, toys and smoke alarms	Zinc	Not classed as hazardous waste – can be disposed with household waste
Zinc chloride	Torches, clocks, shavers, radios, toys and smoke alarms	Zinc	Not classed as hazardous waste – can be disposed with household waste
Alkaline manganese	Personal stereos and radio/cassette players	Manganese	Not classed as hazardous waste – can be disposed with household waste
Primary button cells (i.e. a small flat battery shaped like a 'button' used in small electronic devices)			
Mercuric oxide	Hearing aids, pacemakers and cameras	Mercury	Recycle at council waste facility, if available
Zinc air	Hearing aids, pagers and cameras	Zinc	Recycle at council waste facility, if available
Silver oxide	Calculators, watches and cameras	Silver	Recycle at council waste facility, if available
Lithium	Computers, watches and cameras	Lithium (explosive and flammable)	Recycle at council waste facility, if available
Dry cell rechargeable – secondary batteries			
Nickel cadmium (NiCd)	Mobile phones, cordless power tools, laptop computers, shavers, motorized toys, personal stereos	Cadmium	Recycle at council waste facility, if available
Nickel–metal hydride (NiMH)	Alternative to NiCd batteries, but longer life	Nickel	Recycle at council waste facility, if available
Lithium-ion (Li-ion)	Alternative to NiCd and NiMH batteries, but greater energy storage capacity	Lithium	Recycle at council waste facility, if available

§2.2 Electrical Resistance (Ohm's Law)

- Ohm's Law

$v = i\,R$ [$R =$ the resistance in Ohms (Ω)]



$v = iR$ $\quad$ $v = -iR$

※ Georg Simon Ohm (1789-1854 Germany)

Ohm's Law 1827

https://en.wikipedia.org/wiki/Ohm%27s_law

https://en.wikipedia.org/wiki/Ohm

- the circuit symbol for an 8 Ω resistor

- the conductance G (unit : siemens [S])

$$G = \frac{1}{R} \quad [\mathrm{S}]$$

- Power

$$p = i\,v = i^2 R = \frac{v^2}{R} \quad [\mathrm{Watts}]$$

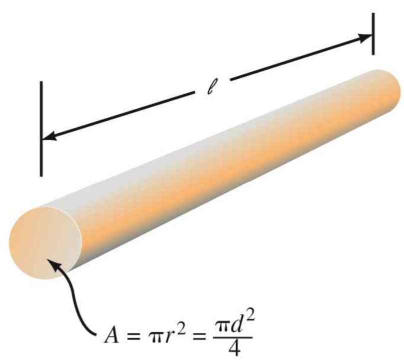
● Resistance

$$R = \rho \frac{l}{A} \quad [\Omega]$$

ρ : resistivity [$\Omega\, m$]

l : length [m]

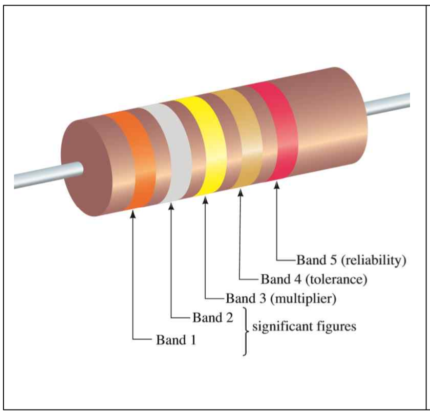
A : cross-sectional area [m^2]



※ Resistivity of Materials, $\rho\ [\Omega m]$

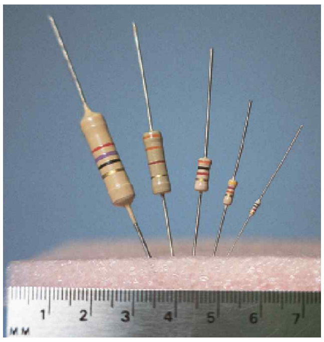
Material	**Resistivity, ρ, at 20° C (Ω-m)**
Silver	1.645×10^{-8}
Copper	1.723×10^{-8}
Gold	2.443×10^{-8}
Aluminum	2.825×10^{-8}
Tungsten	5.485×10^{-8}
Iron	12.30×10^{-8}
Lead	22×10^{-8}
Mercury	95.8×10^{-8}
Nichrome	99.72×10^{-8}
Carbon	3500×10^{-8}
Germanium	20–2300*
Silicon	≅500*
Wood	10^{8}–10^{14}
Glass	10^{10}–10^{14}
Mica	10^{11}–10^{15}
Hard rubber	10^{13}–10^{16}
Amber	5×10^{14}
Sulphur	1×10^{15}
Teflon	1×10^{16}

● Resistor Color Code



Color	Band 1 Sig. Fig.	Band 2 Sig. Fig.	Band 3 Multiplier	Band 4 Tolerance	Band 5 Reliability
Black		0	$10^0 = 1$		
Brown	1	1	$10^1 = 10$		1%
Red	2	2	$10^2 = 100$		0.1%
Orange	3	3	$10^3 = 1\,000$		0.01%
Yellow	4	4	$10^4 = 10\,000$		0.001%
Green	5	5	$10^5 = 100\,000$		
Blue	6	6	$10^6 = 1\,000\,000$		
Violet	7	7	$10^7 = 10\,000\,000$		
Gray	8	8			
White	9	9			
Gold			0.1	5%	
Silver			0.01	10%	
No color				20%	

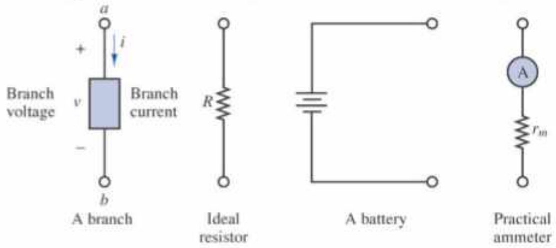
※ Real resistors



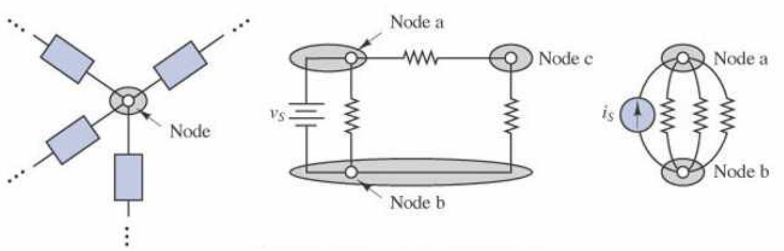
● Terminology

Branch & Node : Basic Units in the Circuit

• Branch : any portion of a circuit with two terminals

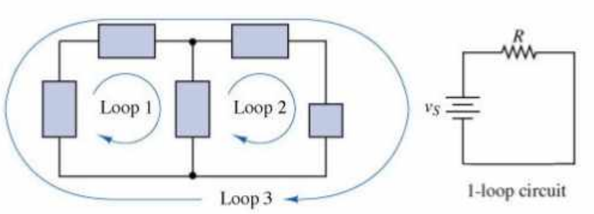


• Node : junction of two or more branches

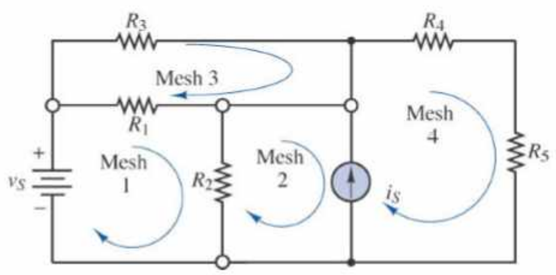


Loop & Mesh : Basic Units in the Circuit

• Loop : any closed connection of branches



• Mesh : a loop that does not contain other loops



Charge & Current

• **(Electric) Charge:**

- A fundamental conserved property of matter (proton and nuclei : +, electron: -), which determines their electromagnetic interaction.
- Quantized by elementary charge (q_e= - 1.602 X 10^{-19}C).
- Electrons have a charge of $-q_e$ & protons have a charge of $+q_e$.
- Unit : coulomb (C)

• **(Electric) Current:**

- Time ("flow") rate of electrical charge passing through a certain area.
- High (low) current: many (few) charge carriers are flowing.
- $i = dq/dt$ (C/s)
- Unit : ampere (A)

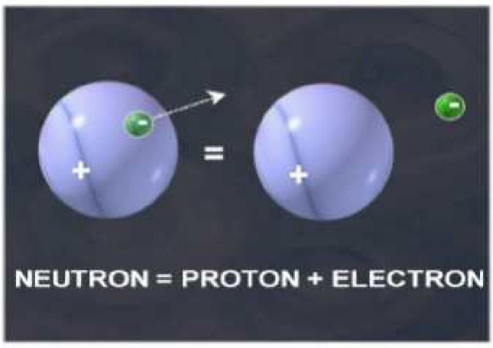


Image adapted from www.osha.gov

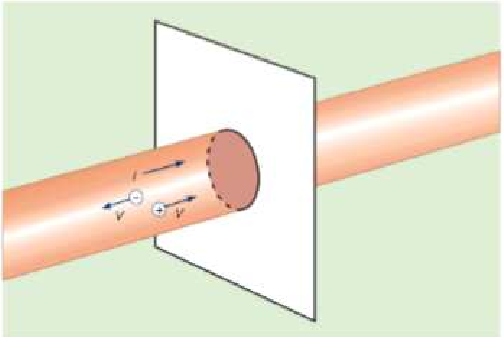
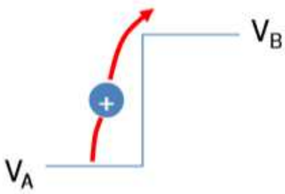


Image adapted from www.britannica.com

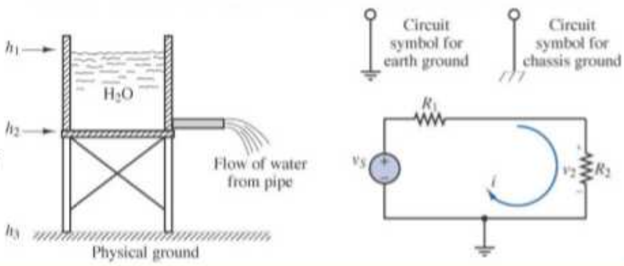
Voltage & Ground

• Voltage (Potential difference) :
- The energy required to move charge from one point to the other.
- Higher voltage means that more energy is required to move charge.
- Unit : Volt (= Joule / Coulomb)



• Reference Voltage:
- In a circuit, any one node may be chosen as the reference node, such that all node voltages may be referenced to this reference voltage.

• Ground (voltage):
- Reference point in an electrical circuit from which other voltages are measured, a common return path for electric current, or a direct physical connection to the earth.

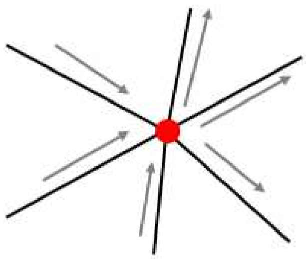


§2.4 Kirchhoff's Laws

- Gustav Kirchhoff(1824-1887 German) first stated in 1848.
  https://en.wikipedia.org/wiki/Gustav_Kirchhoff

Kirchhoff's Current Law (KCL)

- No charge can be created (except for current or voltage sources) in the electrical circuit;
- The sum of the currents at a node must equal zero.
- In other words, sum of "incoming" currents equals that of "outgoing" currents.



$$\sum_{k=1}^{N} I_k = 0$$

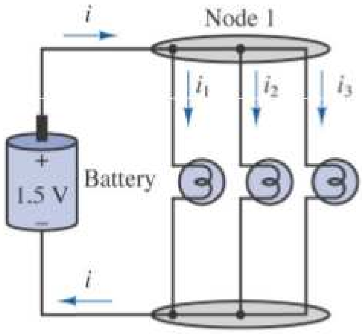
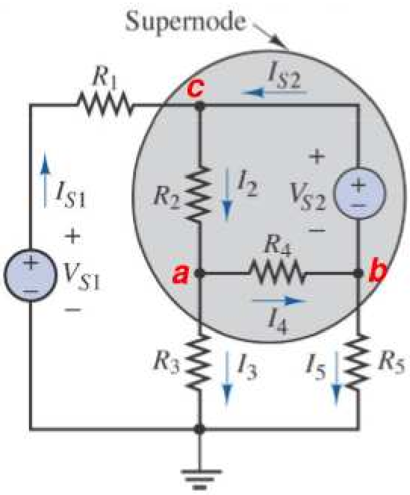


Illustration of KCL at node 1: $-i + i_1 + i_2 + i_3 = 0$

- We can apply the KCL to solve simple electrical network problems.

Problem 2

- Known quantities I_2=3A, I_3=2A, I_5=0A
- Find I_{S1} and I_{S2}.



[Solution]

At node a,
$I_2 - I_3 - I_4 = 0\ A$
Thus,
$I_4 = I_2 - I_3 = 1\ A$

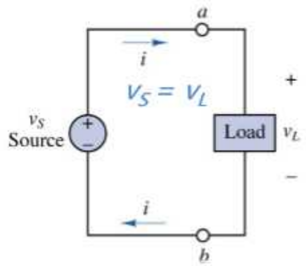
At node b,
$I_4 - I_{S2} - I_5 = 0\ A$

Thus,
$I_{S2} = I_4 - I_5 = 1\text{-}0 = 1\ A$

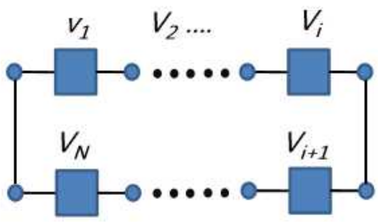
At node c,
$I_{S1} + I_{S2} - I_2 = 0\ A$
Thus,
$I_{S1} = I_2 - I_{S2} = 3\text{-}1 = 2\ A$

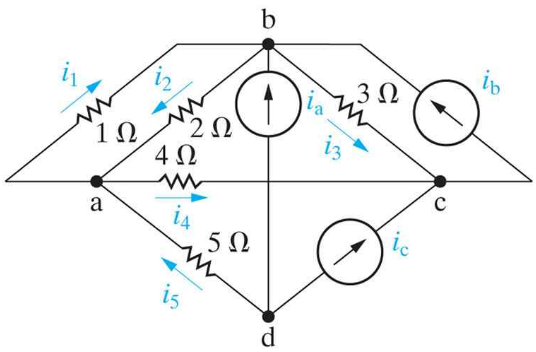
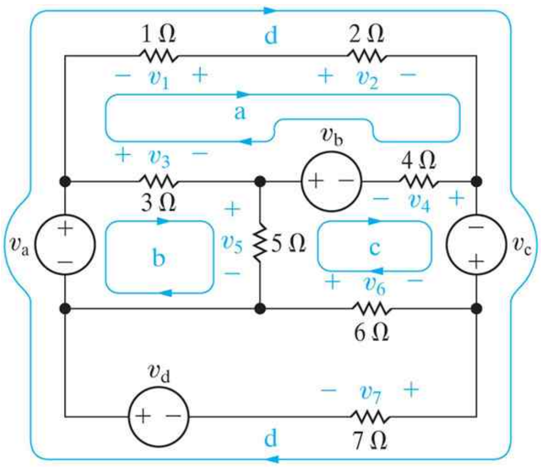
Kirchhoff's Voltage Law (KVL)

- Kirchhoff's Voltage Law (KVL) :
- Net voltage around a close circuit is zero.
- In other words, the sum of all voltages associated with sources equal that of load voltages.
- In equation,

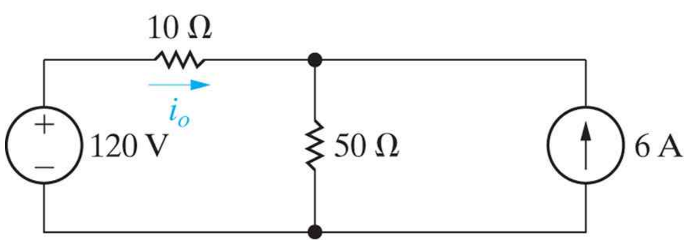


$$\sum_{k=1}^{N} v_k = 0$$

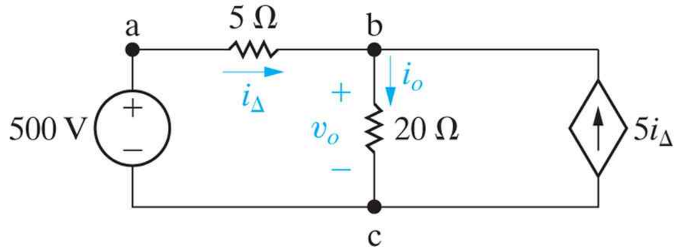


(Example) using KCL	(Example) using KVL
	

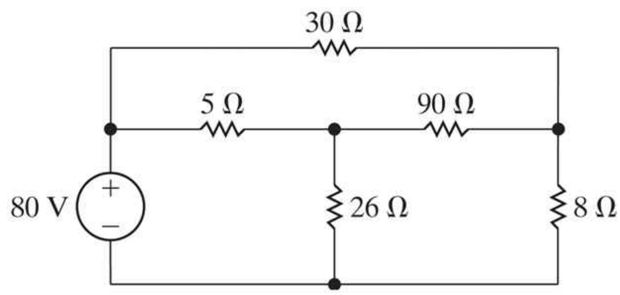
(Example) Applying Ohm's Law, KVL and KCL

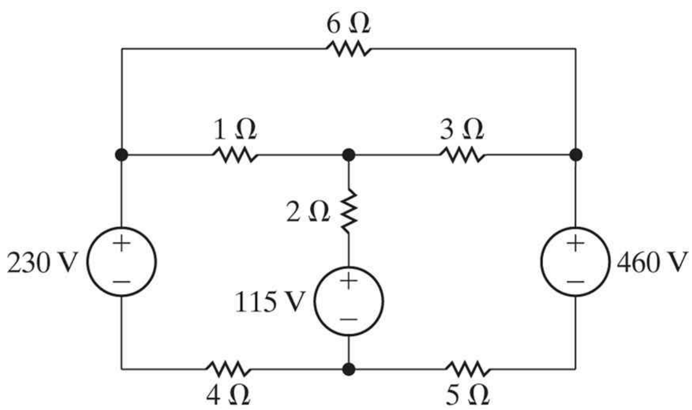


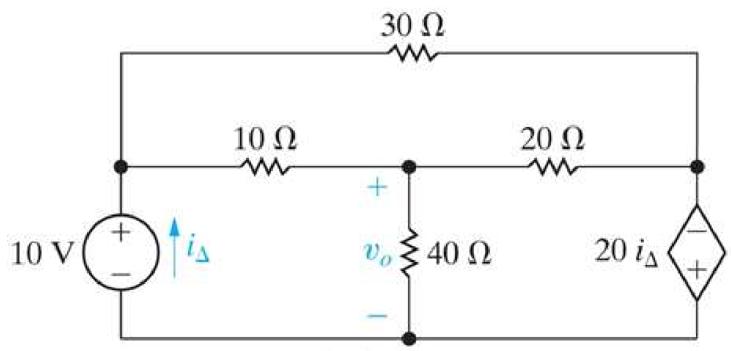
§2.5 Analysis of a Circuit Containing Dependent Sources

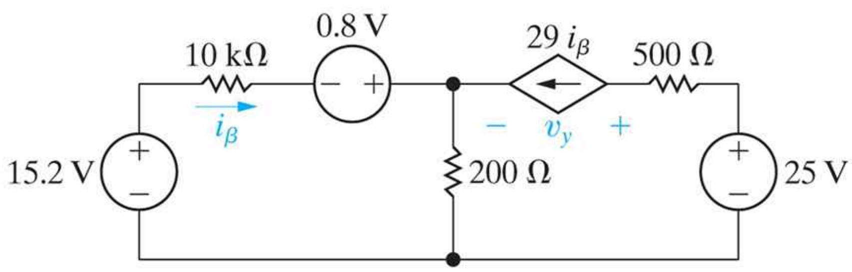


(Circuit Analysis Examples using Ohm's Law, KCL and KVL)









(Examples)

1.34 Find the power that is absorbed or supplied by element 2 in Fig. P1.34.

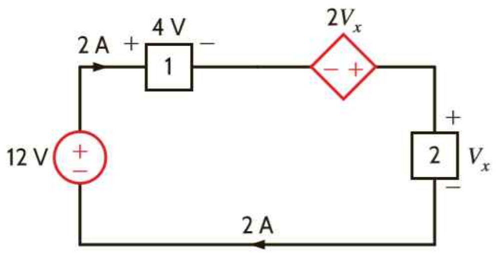


Figure P1.34

1.35 Find I_x in the network in Fig. P1.35.

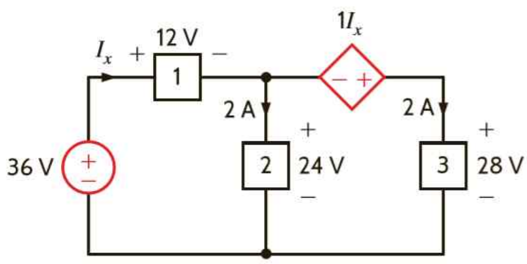


Figure P1.35

1.39 Find the power absorbed or supplied by element 1 in Fig. P1.39.

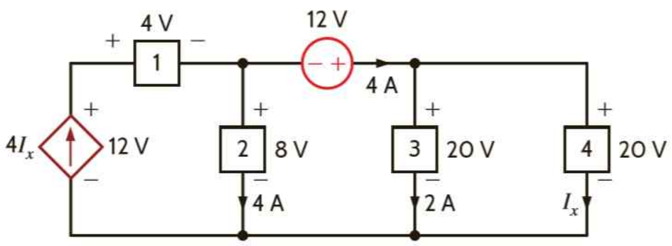
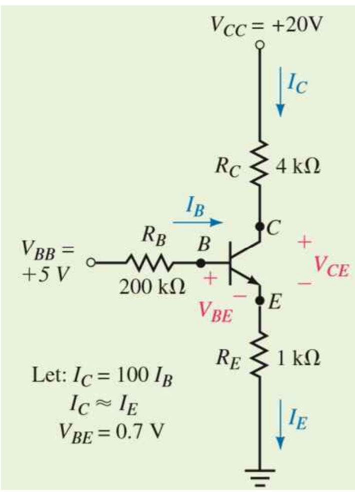


Figure P1.39

(Example) Use the given conditions to determine I_C, I_E , V_{CE} and V_B .



$I_B = 14.3\ \mu\text{A}$

$I_C = 1.43\ \text{mA}$

$I_E = I_C + I_B \cong I_C = 1.43\ \text{mA}$

$V_{CE} = 12.8\ \text{V}$

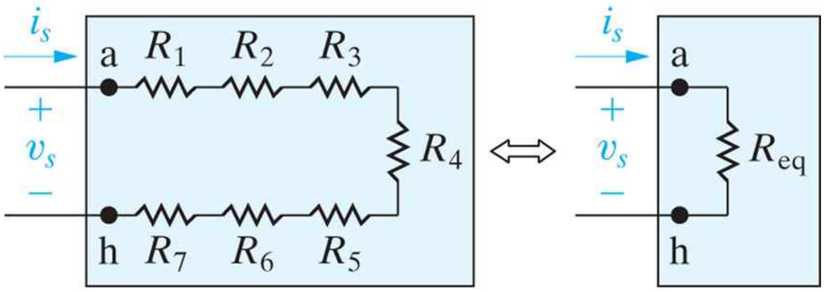
$V_B = 2.13\ \text{V}$

※ An npn BJT (Bipolar Junction Transistor) Amplifier circuit

Chap.3 Simple Resistive Circuits

§3.1 Resistors in Series

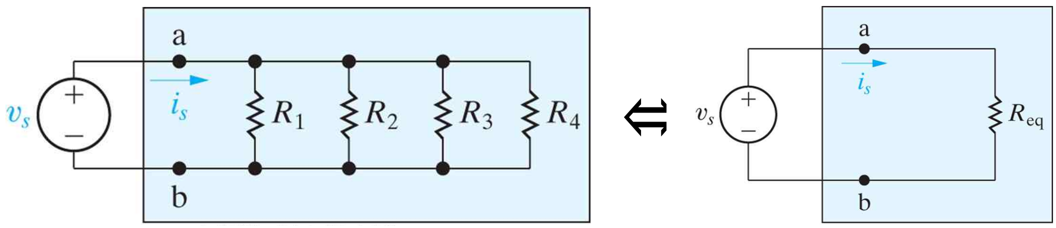
- Combining resistors in series



$$R_{eq} = \sum_{n=1}^{k} R_n = R_1 + R_2 + \cdots + R_k$$

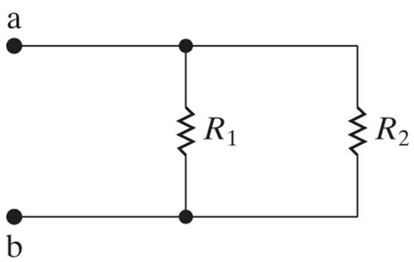
§3.2 Resistors in Parallel

- Combining resistors in parallel



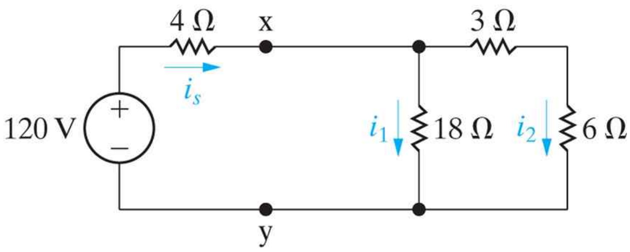
$$\frac{1}{R_{eq}} = \sum_{n=1}^{k} \frac{1}{R_n} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

- Two resistors in parallel (often used)

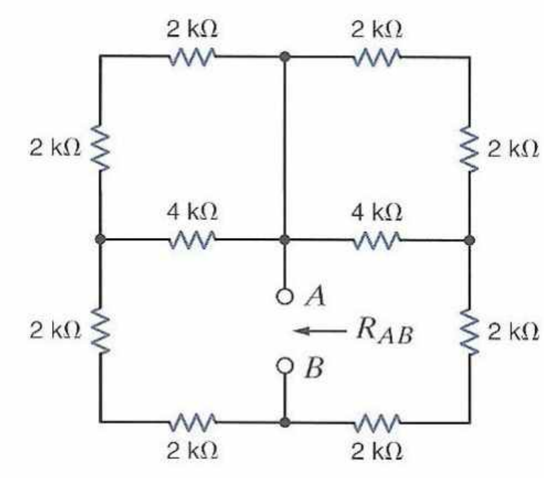


$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

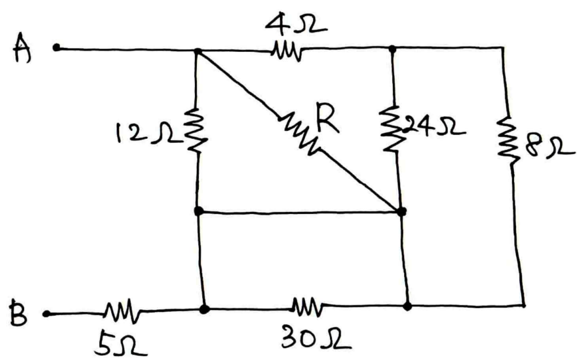
(Example) Find i_s, i_1, and i_2.



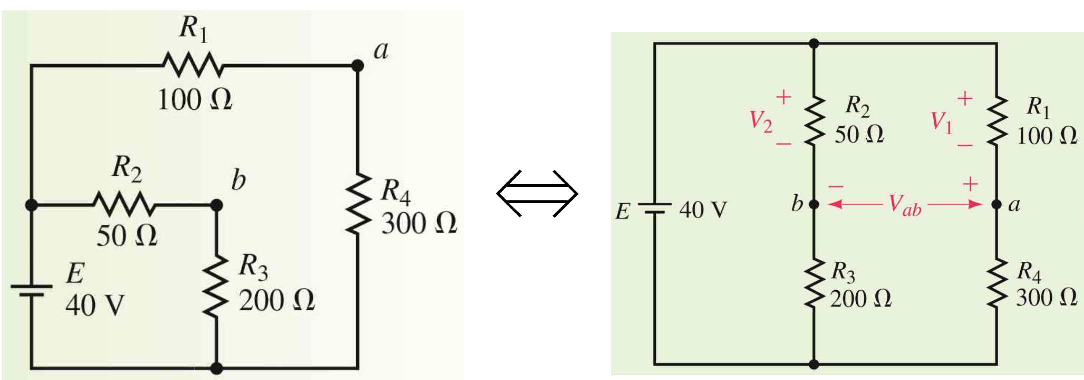
(Example) $R_{AB} = ?$



(Example) If $R_{AB} = 9\,\Omega$, find the value of R.

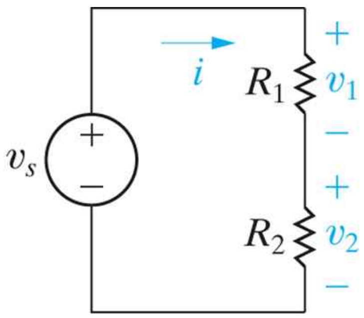


(Example) Find V_{ab} . ($V_{ab} = -2.0\ \mathrm{V}$)



§3.3 The Voltage-Divider and Current-Divider Circuits

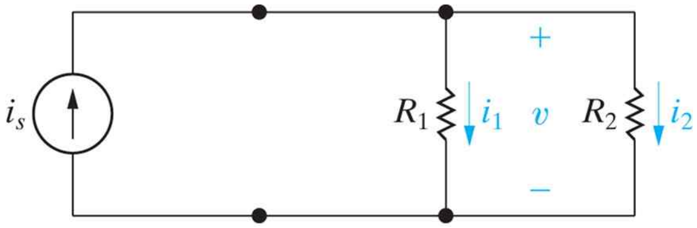
- the voltage divider



$$v_1 = \frac{R_1}{R_1 + R_2} v_s$$

$$v_2 = \frac{R_2}{R_1 + R_2} v_s$$

- the current divider



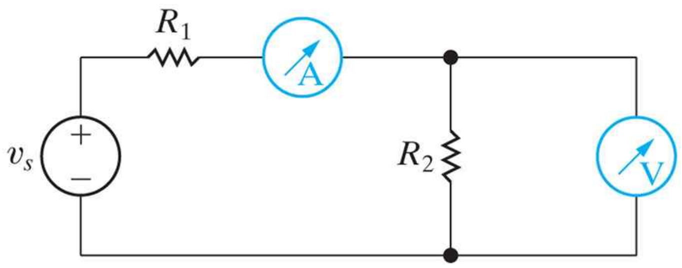
$$v = i_1 R_1 = i_2 R_2 = \frac{R_1 R_2}{R_1 + R_2} i_s$$

$$\therefore \quad i_1 = \frac{R_2}{R_1 + R_2} i_s \,, \qquad i_2 = \frac{R_1}{R_1 + R_2} i_s$$

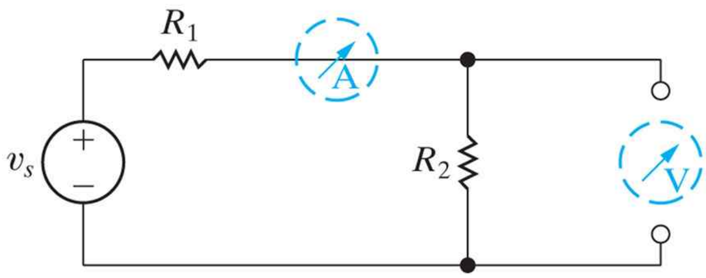
§3.5 Measuring Voltage and Current

- measuring voltage [voltmeter] and current [ammeter]

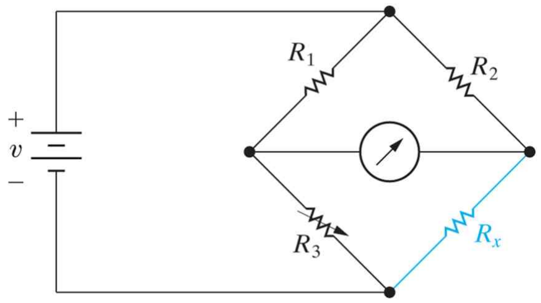
an ammeter connected to measure the current in R_1
a voltmeter connected to measure the voltage across R_2



- a short-circuit model for the ideal ammeter,
  and an open-circuit model for the ideal voltmeter



§3.6 The Wheatstone Bridge

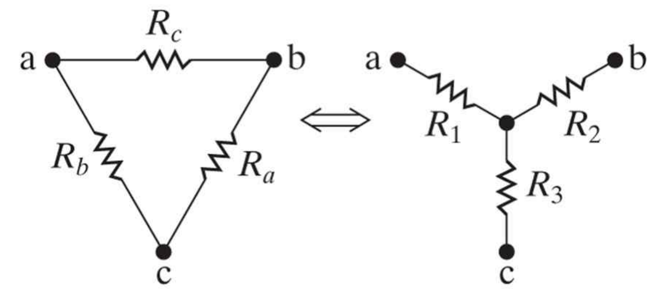


Adjust R_3 until the current through ammeter = 0, then

$$R_x = \frac{R_2}{R_1} R_3$$

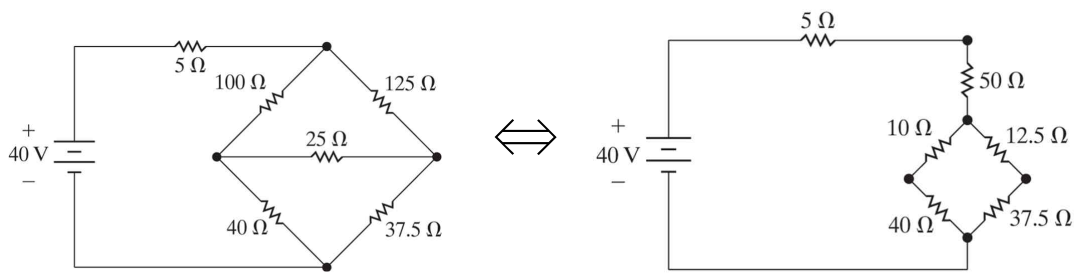
§3.7 Delta-to-Wye (Pi-to-Tee) Equivalent Circuits

- The Δ-to-Y transformation

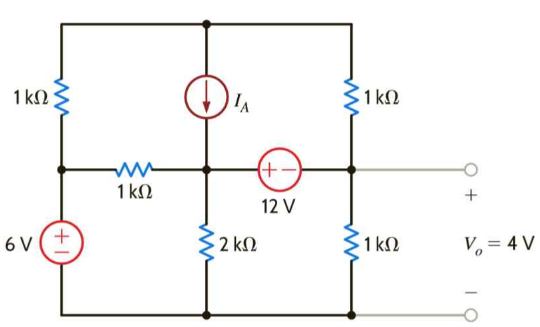
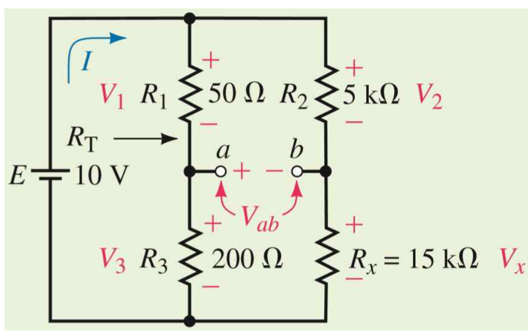


$R_a = \dfrac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$	$R_1 = \dfrac{R_bR_c}{R_a + R_b + R_c}$
$R_b = \dfrac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$	$R_2 = \dfrac{R_cR_a}{R_a + R_b + R_c}$
$R_c = \dfrac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$	$R_3 = \dfrac{R_aR_b}{R_a + R_b + R_c}$

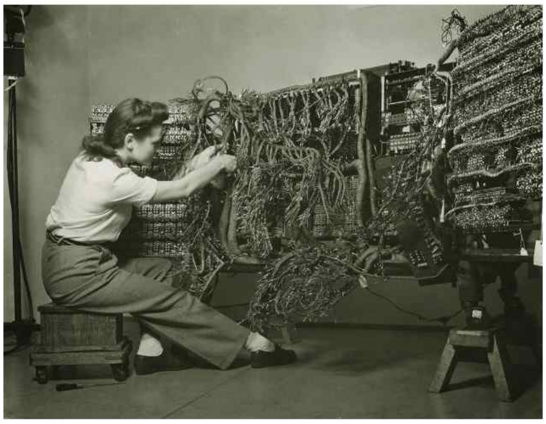
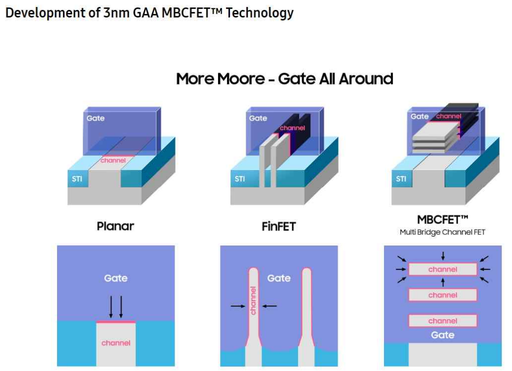
(Example) transformation



(Examples)

Find I_A . $(I_A = 42\ \mathrm{mA})$	Find V_{ab} . $(V_{ab} = 0.5\ \mathrm{V})$
	

(Comparison)

1958 IBM Computer	June 21, 2023 Samsung 3 nm technology
	

https://semiconductor.samsung.com/news-events/tech-blog/3nm-gaa-mbcfet-unrivaled-sram-design-flexibility/

https://www.samsung.com/semiconductor/minisite/exynos/products/all-processors/